

THE SEARCH FOR LIFE

PEOPLE HAVE ALWAYS BEEN FASCINATED BY THE POSSIBILITY OF LIFE BEYOND EARTH, BUT THE ODDS OF ITS EXISTENCE, AND THE PROSPECTS FOR ITS DETECTION, HAVE BEEN GIVEN SEVERAL HUGE BOOSTS IN RECENT YEARS.

The search for life in the Universe has been transformed by the discovery of volcanically active ocean moons in our own Solar System, and countless exoplanets orbiting other stars, offering potential homes for different forms of life.

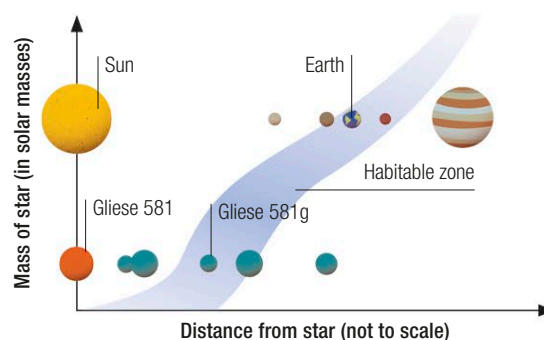
Requirements for life

Traditional theories assumed that life got started in the so-called “primordial soup” of a shallow, warm, chemical-rich seas on the surface of the early Earth. This appears to be the ideal condition to provide the three necessities of life: carbon-based chemicals, water, and an energy source in the form of sunlight. Today, the carbon and water requirements still seem reasonable, since they allow the development of complex chemistry. But the discovery of “extremophiles”—organisms that feed on chemical energy in the pitch darkness of deep-sea volcanic vents, or even deep inside hot subterranean rocks—have changed ideas about what life is, and the conditions it needs to survive.



◁ Transporting life

Some astronomers have speculated that life on Earth might not have needed to have evolved from scratch. Instead, either life itself, or at least complex chemicals that would help it get started, might be transferred between planets inside meteorites or comets.



◁ Goldilocks zone

The most hospitable worlds around any star are likely to be Earth-like planets orbiting within a habitable “Goldilocks zone” around their stars, where temperatures are just right for liquid water to survive on the surface. A few such worlds have been identified, and they are probably abundant in the Milky Way.

▽ Hardy organisms

Tardigrades are tiny animals, also known as “water bears,” whose durability shows how life could persist in conditions very different from those on Earth’s surface. They can survive extremes in temperature and pressure, exposure to the vacuum of space, and bombardment with radiation.

